Associate of Engineering

CCIT4080 Project on Knowledge Products Development

Student Supplementary Resource Guide (2026/27)

Free and Educational Tools for Project Development

AI-Generated Supplementary Document

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About This Guide

This guide provides a curated list of optional external resources that students may explore when planning, designing, implementing, testing, documenting, or presenting a CCIT4080 project. The resources are suggestions only. A group should choose tools because they suit the engineering problem, not because they appear in this guide.

CCIT4080 projects may be hardware-focused, software-focused, integrated hardware-software systems, simulation-oriented, or research-informed development projects. A functional prototype remains the central goal, so the most useful resource will depend on the project scope and the evidence the group needs to produce.

AI-generation notice
This document was generated entirely using generative AI and received only limited manual checking for obvious errors, duplication, course relevance, and readability. Service names, eligibility requirements, quotas, interfaces, licence conditions, and free-tier limits can change. Students should always verify the current information on the official provider website before relying on a service.

GitHub Student Developer Pack as the starting point

The GitHub Student Developer Pack is included as the first resource in this guide because it provides one convenient gateway to many student benefits. Individual services already offered through the Pack - for example, Azure, JetBrains, Notion, MongoDB, Arduino, Deepnote, Heroku and other Pack partners - are generally not duplicated as separate entries below. The remaining resources are intended to complement the Pack. GitHub Student Developer Pack

Before signing up

- Use an official provider website. Avoid unofficial download pages or third-party licence sellers.
- Prepare your College student email and proof of current student status when an education plan requires verification.
- Read the educational-use, non-commercial-use, public-project, storage, quota, and renewal conditions before starting important project work.
- Do not enter payment-card details or upgrade to a paid plan unless the group deliberately decides that a paid service is necessary.
- Do not upload confidential, personal, copyrighted, or sensitive project data to an external service without considering privacy and licence implications.
- When an AI tool or AI-generated output is used in assessed work, follow the current CCIT4080 instructions on declaration, documentation, and academic honesty.

Quick Start - Which Resources Should I Look At?

You do not need to register for everything. Start with the type of work your group actually needs to complete.

PACK 1 **GitHub Student Developer Pack**
One verified student account can unlock a broad set of developer, cloud, AI, learning, testing, security and infrastructure offers.

2 **Electronics, PCB & Embedded Systems**
Altium Student Lab, KiCad, EasyEDA, Wokwi, LTspice, and Ansys Electronics Desktop Student.

3 **CAD, Mechanical Design & Engineering Simulation**
Autodesk Education, Onshape, PTC Creo, and Siemens Solid Edge.

4 **UI/UX, System Design & Collaboration**
Figma for Education, Visual Paradigm Community Edition, Miro Education, Unity Student Plan.

5 **Web, Backend, Cloud, AI & Data Work**
Firebase Spark, Supabase Free, Cloudflare Pages / Workers, Google Colab, MATLAB Online Basic, AWS Educate and IBM SkillsBuild for University Students.

Choose selectively
Using more tools does not make a project better. A smaller, well-justified toolchain that the group understands, tests, documents, and can explain is usually preferable to a large collection of services.

1. GitHub Student Developer Pack

A broad verified-student bundle covering development, cloud, AI, infrastructure, testing, security, learning and related services. It is a useful first place to check before selecting individual project tools.

1. GitHub Student Developer Pack	
Link	https://education.github.com/pack
What to prepare	Create or use a GitHub account; prepare your current College student email and evidence of current enrolment or student status. GitHub Education may request additional verification information or documents. Keep your account name and student records reasonably consistent to reduce verification problems.
What you can find there	A large collection of student benefits centred on software development and related technical work. The current Pack includes GitHub Pro and GitHub Copilot Student, Codespaces, cloud and hosting benefits, professional IDEs, domains, databases, learning platforms, testing/monitoring tools, security tools, IoT services and other partner offers. The exact partners, quotas and durations can change over time.
Suitable CCIT4080 project types	Potentially useful for almost any software-intensive CCIT4080 project: web/mobile applications, AI/data projects, cloud backends, IoT and embedded systems, collaborative coding, source control, deployment, automated testing and technical documentation. Hardware-focused groups can still use GitHub repositories for firmware, PCB files, documentation, issues and version history.
Free-access note	GitHub Student Developer Pack eligibility must be verified and periodically renewed. Each partner benefit has its own activation steps, duration, quotas and licence terms; some require a separate account. Students should read the current Pack page and each provider's conditions before depending on a benefit for a project.

Selected benefits described next

The following 1A-1H entries are summaries of high-value services inside the GitHub Student Developer Pack. Students still activate each benefit through the Pack and must follow the individual provider terms.

1A. GitHub Pro, GitHub Copilot Student and GitHub Codespaces	
Link	GitHub Student Developer Pack
What to prepare	A verified GitHub Education student account. Keep the College email and current proof of enrolment available for renewal or re-verification.
What you can find there	GitHub Pro for source control and collaboration; GitHub Copilot Student for AI-assisted coding; and Codespaces for cloud-hosted development environments. Current AI-credit and usage allowances can change.
Suitable CCIT4080 project types	Almost any software, AI, web, mobile, IoT or embedded project requiring version control, issue tracking, collaborative coding, code review, cloud development or technical documentation.
Free-access note	These are Pack benefits, not separate permanent licences. Review current Copilot, Codespaces and acceptable-use limits, and disclose AI assistance in assessed work as required.

1B. Microsoft Azure	
Link	Activate through the GitHub Student Developer Pack
What to prepare	A verified GitHub Student Developer Pack account and a Microsoft account. The current Pack describes the Azure offer for students aged 18 or above; follow the provider verification steps.
What you can find there	Access to selected Azure cloud services together with the student credit described on the current Pack page. Services can include app hosting, functions, databases, storage, virtual machines, monitoring and AI-related services.
Suitable CCIT4080 project types	Cloud backends, APIs, web/mobile applications, databases, IoT dashboards, serverless functions, AI services and projects requiring temporary cloud computing or public deployment.
Free-access note	Credit, service quotas, eligible regions and expiry conditions apply. Monitor usage and do not enable paid resources unless the group deliberately accepts the billing risk.

1C. JetBrains	
Link	Activate through the GitHub Student Developer Pack
What to prepare	A verified GitHub Student Developer Pack account and a JetBrains account. The student entitlement normally requires periodic renewal while the user remains eligible.
What you can find there	Professional desktop development environments such as IntelliJ IDEA, PyCharm, CLion, WebStorm, Rider, DataGrip and related JetBrains tools, depending on the current education offer.
Suitable CCIT4080 project types	Python, Java, C/C++, web, .NET, database and cross-platform software projects where code navigation, debugging, testing and professional IDE features are useful.
Free-access note	The licence is for educational use and is renewable subject to eligibility. Check the current non-commercial and graduation/expiry conditions before relying on it after the course.

1D. Notion	
Link	Activate through the GitHub Student Developer Pack
What to prepare	A verified GitHub Student Developer Pack account and a Notion account. Use a College email where practical and review the workspace-sharing settings before adding group content.
What you can find there	The education/Plus-style workspace benefit described by the Pack, with project pages, databases, task tracking, shared notes, longer history and additional AI responses under the current offer.
Suitable CCIT4080 project types	Any group needing a shared workspace for requirements, meeting notes, task allocation, risk registers, design decisions, test records, references or a supplementary project log.
Free-access note	Notion is optional and does not replace required SOUL submissions or the formal Logbook. AI-generated content must be checked, documented and declared where required.

1E. MongoDB	
Link	Activate through the GitHub Student Developer Pack
What to prepare	A verified GitHub Student Developer Pack account and a MongoDB account. Plan the collection structure, access controls, region and backup/export approach before storing project data.
What you can find there	The current Pack describes MongoDB Atlas credit, MongoDB Compass, MongoDB University learning resources and a certification benefit. Atlas provides a managed document database in the cloud.
Suitable CCIT4080 project types	Web/mobile applications, IoT data stores, event logs, content systems, dashboards and prototypes that benefit from a document-oriented database rather than a relational schema.
Free-access note	Pack credit is limited and can expire. Track consumption, protect credentials, avoid personal/sensitive data, and keep exportable backups so the project can continue after the credit ends.

1F. Arduino Cloud	
Link	Activate through the GitHub Student Developer Pack
What to prepare	A verified GitHub Student Developer Pack account, an Arduino account and compatible hardware for physical deployment. Browser/device-agent installation may be required.
What you can find there	The current Pack lists time-limited Arduino Cloud access and selected hardware discounts. Arduino Cloud supports connected Things, dashboards, variables, remote monitoring and IoT-oriented workflows.
Suitable CCIT4080 project types	Arduino-based IoT devices, remote sensor monitoring, cloud dashboards, simple device control and prototypes that need a managed connection between hardware and a browser/mobile interface.
Free-access note	The Pack offer is time-limited and does not provide all hardware free of charge. Check supported boards, cloud quotas and what happens when the promotional period ends.

1G. Deepnote	
Link	Activate through the GitHub Student Developer Pack
What to prepare	A verified GitHub Student Developer Pack account and a Deepnote account. Decide what data can be uploaded to an external notebook service and keep local/Git backups.
What you can find there	A collaborative, Jupyter-compatible cloud notebook environment. The current Pack describes access to the Deepnote Team plan with team projects, basic cloud machines, revision history and selected integrations.
Suitable CCIT4080 project types	Collaborative AI/ML, data analysis, signal processing, exploratory research, database-connected notebooks and groups that need shared executable notebooks rather than individual local files.
Free-access note	Cloud-machine, storage, history and integration limits can change. Export notebooks and important data regularly; do not treat the service as the only project archive.

1H. Heroku	
Link	Activate through the GitHub Student Developer Pack
What to prepare	A verified GitHub Student Developer Pack account and a Heroku account. Read the current billing/credit activation steps before deploying applications or add-ons.
What you can find there	A managed platform for deploying and operating web applications, APIs and background processes. The current Pack lists a monthly student credit for a limited period.
Suitable CCIT4080 project types	Flask, Django, Node.js and other web/API prototypes; demonstration backends; scheduled jobs; and projects needing a simple public deployment workflow.
Free-access note	The Pack currently states a fixed monthly credit for 24 months. Services above the credit can incur charges, so monitor usage and remove unneeded deployments after assessment.

2. Electronics, PCB & Embedded Systems

Resources for circuit design, PCB development, electronic simulation, microcontroller prototyping, RF/electromagnetic work, and embedded-system development.

2. Altium Student Lab	
Link	https://www.altium.com/education/students
What to prepare	Academic email address. If automatic verification is unavailable, be ready to provide a valid student card and/or current-semester enrolment proof. A compatible Windows PC is normally required for Altium Designer.
What you can find there	Professional schematic capture and PCB design using Altium Designer, together with Altium 365 collaboration, component libraries, PCB learning materials, and student-oriented training/certification resources.
Suitable CCIT4080 project types	Hardware-centric projects requiring a custom PCB; embedded/IoT devices; sensor interfaces; power/control boards; mixed hardware-software prototypes where a professional PCB workflow is useful.
Free-access note	The student licence is for educational use and is normally renewable while the user remains eligible. Verify the current licence conditions before use.

3. KiCad	
Link	https://www.kicad.org/
What to prepare	A Windows, macOS, or Linux computer. No student email or student verification is required.
What you can find there	Open-source electronic design automation (EDA) software for schematic capture, PCB layout, Gerber/manufacturing outputs, 3D PCB viewing, and integrated circuit simulation workflows.
Suitable CCIT4080 project types	Custom PCB projects, microcontroller boards, sensor/actuator interfaces, analog/digital circuits, power electronics prototypes, and groups that prefer a fully free local toolchain.
Free-access note	Free and open source. It is a strong choice when the group wants to avoid an education-only commercial licence.

4. EasyEDA	
Link	https://easyeda.com/page/pricing
What to prepare	A web browser or EasyEDA desktop client. Create a free account if you want to save, synchronise, or share projects in the cloud. No student verification is required for the free core plan.
What you can find there	Browser/desktop schematic and PCB design environment with simulation support, PCB layout, component libraries, and manufacturing-output tools. It is closely integrated with common PCB prototyping workflows.
Suitable CCIT4080 project types	Entry-level PCB and electronics projects, IoT devices, small sensor/controller boards, rapid PCB prototyping, and groups that want a browser-based EDA workflow.
Free-access note	The core editor has a free plan. Check current storage, cloud, collaboration, and education-feature limits before depending on them.

5. Wokwi	
Link	https://wokwi.com/pricing
What to prepare	A modern web browser. Create an account to save/share projects. Free Community projects would be treated as public unless the current plan states otherwise.
What you can find there	Online simulation environment for Arduino, ESP32, Raspberry Pi Pico and other embedded platforms. Students can test firmware, virtual sensors/displays, wiring logic, and some network-connected behaviour before physical hardware is complete.
Suitable CCIT4080 project types	Arduino/ESP32/Pico projects; embedded firmware; IoT proof-of-concepts; sensor/actuator logic; early-stage prototyping when hardware is not yet available or when repeatable simulation is useful.
Free-access note	The no-cost Community plan is best for public/open projects. Private or advanced features require a paid plan.

6. LTspice	
Link	https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html
What to prepare	A compatible Windows or macOS computer. No student account or education verification is normally required.
What you can find there	Analog Devices' free SPICE simulator with schematic capture and waveform analysis. It is widely useful for analysing analog, power, filter, amplifier, switching, and mixed-signal circuit behaviour before construction.
Suitable CCIT4080 project types	Analog electronics, power supplies, amplifiers, filters, sensor front-ends, switching circuits, signal-conditioning projects, and any project that needs circuit-level simulation evidence.
Free-access note	Free software. Simulation models and real hardware can differ, so measured prototype results should still be used for final validation.

7. Ansys Electronics Desktop Student	
Link	https://www.ansys.com/academic/students/ansys-electronics-desktop-student
What to prepare	A compatible Windows PC with sufficient RAM/storage and administrator rights for installation. Internet access is needed for the download. Student software is intended for educational use.
What you can find there	Student edition of Ansys Electronics Desktop with tools such as HFSS, Maxwell, Q3D Extractor, Icepak, Twin Builder and Circuit for electromagnetic, RF, antenna, electrical-machine, PCB/interconnect, circuit and thermal studies.
Suitable CCIT4080 project types	RF/antenna projects, electromagnetic sensing, motors/actuators, magnetic systems, high-frequency interconnects, electronics thermal analysis, and simulation-heavy electrical-engineering projects.
Free-access note	Student editions have model/problem-size and licensing limits. Check the current Ansys Student product restrictions before choosing a project scope.

3. CAD, Mechanical Design & Engineering Simulation

Resources for enclosure design, mechanical assemblies, 3D-printable parts, structural/thermal/flow analysis, product design, and integrated engineering prototypes.

8. Autodesk Education	
Link	https://www.autodesk.com/education/edu-software/overview
What to prepare	An Autodesk account, current student status, school details, and proof of eligibility if requested by the verification process. Use your institutional email where possible.
What you can find there	Educational access to Autodesk professional software. Depending on current availability, students can use products such as AutoCAD, Fusion and Inventor for CAD, mechanical design, modelling, assemblies, fabrication preparation and related engineering work.
Suitable CCIT4080 project types	Mechanical/electromechanical prototypes, enclosures, robot frames, fixtures, 3D-printed parts, product-design projects, and hardware projects requiring clear engineering drawings or CAD models.
Free-access note	Education access is normally time-limited and renewable while eligible, and is restricted to educational purposes. Verify the exact products and conditions shown in your account.

9. Onshape Student Plan	
Link	https://www.onshape.com/en/education/plans
What to prepare	Create an Onshape account and apply under the Student Plan. Be ready to confirm current student status; use your institutional email where possible. A modern browser is sufficient for normal use.
What you can find there	Professional cloud CAD that runs in a web browser, with parametric part design, assemblies, drawings and collaboration. It is convenient when group members need to work on the same design without managing desktop CAD files manually.
Suitable CCIT4080 project types	Mechanical design, robot/device enclosures, 3D printing, assemblies, fixtures, product-design projects, and cross-device/group collaboration on CAD models.
Free-access note	Education/student plans are for eligible academic users. Check current document privacy, export, simulation, and renewal conditions.

10. PTC Creo - Education	
Link	https://www.ptc.com/en/education/free-software
What to prepare	A PTC account and current student/academic status. Use a school email if available. A compatible computer is required for the Creo installation.
What you can find there	PTC provides Creo education access for college/university learners. Creo supports professional parametric 3D CAD, assemblies, drawings and product-engineering workflows.
Suitable CCIT4080 project types	Mechanical/product-design projects, complex assemblies, enclosure/chassis design, mechanisms, fixtures, and engineering projects where professional CAD practice is part of the learning goal.
Free-access note	Use is subject to PTC education licensing. Verify supported versions, hardware requirements, file compatibility and current eligibility.

11. Siemens Solid Edge Student Edition	
Link	https://solidedge.siemens.com/en/solutions/users/students/
What to prepare	An email/Siemens account, current student or active-learner status, and a compatible Windows computer.
What you can find there	Student edition of Siemens Solid Edge for professional 3D CAD and engineering design. Siemens also provides related student learning resources and, separately, student access to selected CAM learning tools.
Suitable CCIT4080 project types	Mechanical design, assemblies, manufacturing-oriented projects, enclosures, mechanisms, 3D-printable parts, and projects that may benefit from a CAD-to-manufacturing workflow.
Free-access note	Student-edition files and drawings can have compatibility/watermark restrictions relative to commercial Solid Edge. Check these conditions before choosing it as the group's long-term file format.

4. UI/UX, System Design, Collaboration & Interactive Development

Resources for interface prototyping, system/requirements diagrams, group brainstorming, process mapping, and interactive 3D application development.

12. Figma for Education	
Link	https://www.figma.com/education/higher-education/
What to prepare	A Figma account, student email, and higher-education verification. Follow the current Figma education eligibility process.
What you can find there	Professional UI/UX design and collaborative prototyping. Students can build app/web mock-ups, design systems, clickable prototypes, interface flows and shared design artefacts before implementation.
Suitable CCIT4080 project types	Mobile apps, web applications, dashboards, accessibility-focused interfaces, touchscreen/embedded UI, human-computer interaction projects, and any project that benefits from early user-interface testing.
Free-access note	Education entitlements and AI/Dev Mode allowances can change. Verify the current plan features and any AI-use conditions.

13. Visual Paradigm Community Edition	
Link	https://www.visual-paradigm.com/editions/community/
What to prepare	Download/install the Community Edition. No student verification is required. Use it only within the current non-commercial/community licence conditions.
What you can find there	Modelling and diagramming environment supporting UML, use-case and class diagrams, ERD/database diagrams, flowcharts, requirement-related diagrams, network/system diagrams and other software/system-design artefacts.
Suitable CCIT4080 project types	Software systems, web/mobile apps, database-backed projects, IoT architectures, system engineering, requirement modelling, workflow/state modelling, and projects needing professional diagrams in reports.
Free-access note	Community Edition is free for non-commercial use including education, but some export/advanced functions can be limited or watermarked. Check current edition differences.

14. Miro Education	
Link	https://help.miro.com/hc/en-us/articles/360017730473-Education-plan
What to prepare	A Miro account and evidence of current student/education status for the application. Keep a copy of suitable proof in case the provider requests it.
What you can find there	Collaborative online whiteboard with private boards, brainstorming, diagrams, voting, timers and visual project-planning tools. It is useful for turning meetings into visible decisions, flows and action items.
Suitable CCIT4080 project types	Any team project: brainstorming, requirements discovery, user journeys, architecture planning, risk analysis, meeting facilitation, design reviews, test-planning workshops and group coordination.
Free-access note	Student education plans are normally time-limited and team-size limits apply. Verify current duration and collaboration limits before relying on it for the full project.

15. Unity Student Plan	
Link	https://unity.com/products/unity-student
What to prepare	A Unity ID, an email address, and student verification through the current eligibility process (such as SheerID). Have proof of enrolment ready if requested. A capable Windows/macOS computer is recommended.
What you can find there	Student access to the Unity Pro editor and selected education benefits. Unity is a real-time 2D/3D development platform for games, simulations, AR/VR, visualisation and interactive applications.
Suitable CCIT4080 project types	AR/VR projects, educational games, simulation interfaces, interactive 3D visualisation, digital twins/prototypes, spatial user interfaces, and projects where real-time 3D interaction is central.
Free-access note	Unity plan terms, included assets/services, revenue/commercial conditions and renewal rules can change. Verify the current Student Plan terms before publishing externally.

5. Web, Backend, Cloud, AI & Data Work

Resources for app backends, databases, lightweight hosting, serverless functions, notebooks, numerical analysis, cloud learning, and professional technology upskilling.

16. Firebase - Spark Plan	
Link	https://firebase.google.com/pricing
What to prepare	A Google account and a project name/ID. The Spark plan can be started without a payment method. No student verification is required.
What you can find there	Google's app-development backend with services such as authentication, database/storage options, hosting-related features, analytics, crash reporting, messaging and other mobile/web development tools, subject to free-tier quotas.
Suitable CCIT4080 project types	Mobile/web apps, IoT dashboards, login/authentication, simple real-time data, notification-enabled systems, prototypes that need a managed backend without building every server component from scratch.
Free-access note	Some Firebase products/usage require billing or move beyond free quotas. Do not enable a paid billing plan unless the group understands the possible charges.

17. Supabase Free	
Link	https://supabase.com/pricing
What to prepare	Create a Supabase account. No student status is required. Plan the database schema and monitor the current free-project, storage, database, authentication and egress limits.
What you can find there	Managed PostgreSQL backend with database, authentication, APIs, storage, real-time features and developer tooling. It is useful when a student project needs a conventional relational database with a relatively direct development experience.
Suitable CCIT4080 project types	Web/mobile applications, dashboards, user accounts, CRUD systems, IoT data portals, project management tools, database-backed prototypes and software projects requiring SQL/PostgreSQL.
Free-access note	The free plan has project/resource limits and inactive free projects may pause. Keep backups/exportable data and check current quotas.

18. Cloudflare Pages / Workers	
Link	https://developers.cloudflare.com/workers/platform/pricing/
What to prepare	A Cloudflare account. No student verification is required. A Git repository is helpful for common Pages workflows but is not the only deployment route.
What you can find there	Cloudflare Pages can host static/front-end sites, while Workers provides lightweight serverless code at the edge. Together they can support project websites, APIs, prototypes and small web services within the free limits.
Suitable CCIT4080 project types	Project showcase websites, static front-ends, lightweight APIs, serverless prototypes, web dashboards, webhook/end-point experiments, and software projects needing public deployment.
Free-access note	Workers and related products have request/CPU/storage limits. Check the current Free-plan quotas and avoid accidentally enabling paid usage.

19. Google Colab	
Link	https://colab.research.google.com/
What to prepare	A Google account. No student verification is required. Keep your code/data backed up and do not assume that a particular GPU/TPU or session duration will always be available.
What you can find there	Hosted Jupyter Notebook environment that runs Python in the browser with no local setup. Free users can access dynamically allocated compute resources, including accelerator hardware when available.
Suitable CCIT4080 project types	AI/ML experiments, computer vision, data analysis, signal processing, Python prototypes, model evaluation, numerical experiments and projects where students need notebook-based reproducible evidence.
Free-access note	Free compute is not guaranteed and limits can change dynamically. It should not be the only place where important code/data is stored.

20. MATLAB Online Basic	
Link	https://www.mathworks.com/products/matlab-online/matlab-online-versions.html
What to prepare	A free MathWorks account and a modern browser. No student verification is required for MATLAB Online Basic. Monitor the monthly usage allowance.
What you can find there	Browser-based MATLAB with a monthly free usage allowance, selected commonly used products, MATLAB Drive storage, and access to core numerical, plotting and engineering-computation workflows.
Suitable CCIT4080 project types	Signal processing, numerical modelling, control/matrix calculations, data analysis, algorithm prototyping, plotting, engineering mathematics, and projects where MATLAB-style analysis is useful but no campus licence is assumed.
Free-access note	The Basic version has monthly-hour, storage and product limits. Confirm the current included products before designing a workflow around a specialised toolbox.

21. AWS Educate	
Link	https://aws.amazon.com/education/awseducate/
What to prepare	An email address. AWS Educate can be joined without a credit card. No separate GitHub Student Pack membership is required.
What you can find there	Free cloud-learning pathway with beginner-friendly training and hands-on labs covering AWS concepts such as storage, compute, networking, databases, cloud operations and AI-related topics.
Suitable CCIT4080 project types	Students who need to learn cloud architecture before implementing a web/cloud component; cloud-oriented research; API/backend planning; projects where understanding deployment concepts is more important than receiving production hosting credit.
Free-access note	Treat AWS Educate mainly as a learning/lab resource. It is not equivalent to an unrestricted free AWS production account or a guaranteed pool of deployment credits.

22. IBM SkillsBuild for University Students	
Link	https://skillsbuild.org/university/students
What to prepare	An email address and an IBM SkillsBuild account. Use current university/student information where requested. No payment is required for the university-student learning resources.
What you can find there	Free learning platform covering technology and professional skills, including AI, data, cybersecurity and related areas, with structured learning paths and selected digital credentials.
Suitable CCIT4080 project types	Projects where students need to acquire unfamiliar AI/data/cybersecurity knowledge, strengthen background study, understand professional workflows, or build supporting skills before implementing the project.
Free-access note	This is primarily a learning and credential resource, not a project hosting/deployment platform.

Final Reminder - Use the Resource, Do Not Let the Resource Define the Project

A CCIT4080 project should begin with a problem, objectives, requirements, engineering decisions, implementation and testing. A free tool is useful only when it helps the group carry out those steps more effectively.

Free does not always mean unrestricted

Education licences may be limited to non-commercial use; cloud tiers may have quotas; free projects may be public; student editions may limit model size, export, file compatibility, or collaboration; and eligibility may need to be renewed. Keep project files backed up in formats that can be moved to another tool if necessary.

A sensible selection process

- Start with the project requirements and identify the specific task that needs a tool.
- Compare two or more approaches when the tool choice affects architecture, performance, cost, privacy, maintainability, or project risk.
- Prefer tools the group can explain and use independently rather than tools that merely produce attractive outputs.
- Record important versions, settings, assumptions and licences in the Logbook or technical documentation when they affect reproducibility.
- Keep original source files, raw data, test evidence and exportable backups throughout the project.
- For assessed work, explain what the group actually designed, implemented and tested. A commercial or AI-assisted tool does not substitute for engineering evidence.

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